CO2 - AT1 - ANALYTICAL PROBLEM SOLVING

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COURSE NAME : DATA STRUCTURES

COURSE CODE : CSA0307

1) Given a singly linked list L: L1 → L2 → L3 .. → Ln-1 → Ln, reorder and print it to : L1 → Ln → L2 → Ln-1

Sol: Given:

L1 → L2 → L3 → ... → Ln-1 → Ln

Reordered List:

L1 → Ln → L2 → Ln-1 → L3 → Ln-2 → ...

Code:

```
#include <stdio.h>
int main () {
    int a[100], n, i;
    scanf ("%d", &n);
for (i=0; i<n ; i++)
    Scanf ("%d", &a [i]);
for (i=0; i<n/2; i++)
    Printf ("%d %d", a[i], a[n-1-i]);
if (n % 2)
    Print f("%d", a[n/2]);
return 0;
}
```

Input:

5

1 2 3 4 5

Output:

1 5 2 4 3

Insert the following elements 150, 250, 350, 450, 550 into the double Linked list and delete the elements 150 and 250 from the double linked list. show the linked structure using a node diagram before and after delete operations.

Doubly Linked List

Before Deletion

Insert: 150, 250, 350, 450, 550

NULL

↑

[150] ⇌ [250] ⇌ [350] ⇌ [450] ⇌ [550]

Delete 150 & 250

NULL

↑

[350] ⇌ [450] ⇌ [550] ↓

NULL

Final linked list : NULL ← 350 ⇌ 450 ⇌ 550 → NULL